

# **Does modernization impact democratization?**

## **Introduction**

Democracies are the most popular form of government in the modern world, with 96 out of 167 countries being classified as democracies of varying types in 2017 (Desilver, 2019), which could be explained by the normative acceptance of democracies as a regime type (Sen, 2004, p.9). The value-driven appeal of democracies is further strengthened by econometric evidence which suggests that the average rate of investment, labor productivity and efficient allocations of investments were all higher in democracies in comparison to dictatorships (Przeworski, 1996, p.40). While such evidence suggests that economic modernization could be causally linked to democracy, there remains a large debate within the academic community regarding the nature of the relationship between the two. One side of the debate argues that there is no causal effect of income on democracy (Acemoglu, 2008), while the other side argues that income does matter positively for democratization (Boix, 2009). Hence, this technical report aims to answer the question ‘Does modernization impact democratization?’ with the aim of assessing causal claims between the two variables. This report will aim to achieve data triangulation by utilizing four different quantitative techniques- Ordinary Least Squares regression, Mediation Analysis, Matching and Instrumental Variables, to improve the statistical strength of the conclusions drawn.

## **Causal Question**

Does modernization impact democratization?

Hypothesis- Modernization, measured via income per capita, has a positive causal impact on democratization.

This hypothesis is made in alignment with classical modernization literature, where higher income per capita is associated with urbanization and expanded access to education (Goorha, 2017), which ultimately creates a strong middle class. With the existence of a well-positioned middle class, societal facets of inequality and bureaucratic corruption are more likely to decrease, and the middle class will be better incentivized to fight for a say (Chong, 2015) and

be positioned to participate politically. This increases the likelihood of democracy to emerge in a modernized society.

### **Causal identification**

In order to appropriately address the causal question at hand, and work towards a statistically supported answer, certain causal threats to the question must be pre-emptively identified. The first causal threat inherent to the question remains reverse causality, as while modernization may cause democratization, it has also been argued that democracies cause modernization instead (Acemoglu, 2019, p.47). The second causal threat identified is the potential for omitted variable bias to exist, as within this report, modernization is represented by GDP per capita, which in itself will not account for other factors that impact democratization. Finally, the question suffers from selection bias, as politically stable countries may be predisposed to higher incomes and democratization, which would result in unfair comparisons and biased assessments and may result in correlations drawn being skewed.

Previous studies that have undertaken research about this question, such as Boix. et al (2009), dealt with these causal threats by utilizing lagged income in his calculations, adding structural controls, and by explicitly modelling selection. By employing lagged income during his estimations of the impact of per capita income on democracy, Boix addresses the problem of reverse causality as a country that is categorized as a democracy in the future, cannot have any causal impact on the income per capita of the past, thereby restricting the causal pathway investigated to be unidirectional. However, it could be argued that the lagged per capita income was influenced by the political regime of the past, hence mitigating reverse causality for analysis only at a short-term level.

Boix accounts for the omitted variable bias by running models with controls for factors such as income inequality, and whether the country was occupied by authoritarian regimes such as the Soviet Union in the past. By doing so, Boix is able to investigate the causal impact income has on democracy, by ensuring other factors that could also impact democratization don't inflate the statistics. However, despite controlling for these variables, Boix would not be able to control for unobserved cultural and historical factors that would impact democratization, hence not being able to completely account for omitted variables.

Lastly, Boix attempts to counter selection bias by modelling authoritarian countries and their probability to democratize or face democratic breakdown over a period of time, hence making the comparisons fairer. However, this method does not account for survivorship bias, with the fact that countries that survive long enough to be able to democratize are already a selected group within themselves, hence while this method reduces selection bias, it is not able to completely eliminate it.

With the antecedent literature employing such methods to investigate the causal relationship between modernization and democratization better, this report will aim to accomplish the same using 4 identification strategies- Ordinary Least Squares regression, mediation analysis, instrumental variables, and matching. To begin with, Ordinary Least Squares regression will be employed to investigate the correlation seen between income per capita and modernization and will work to reduce concerns related to omitted variable bias by having the regression be conducted with the addition of controls. Following this, mediation analysis will be utilized to understand one of the causal mechanisms through which economic modernization and income per capita affect democratization. An instrumental variable approach will be undertaken after, to address reverse causality and ensure that the causal relationship between income per capita and democratization operates in the established direction. Finally, matching will be used to test for the robustness of the relationship between the two variables under conditions which reduce selection bias.

## **Data**

For the purpose of all the regressions and statistical analysis conducted, data was taken from The Quality of Government Standard Dataset 2025<sup>1</sup>, within which the Authoritarian Regimes

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<sup>1</sup> Quality of Government (2024). *Data Downloads*. [online] University of Gothenburg. Available at: <https://www.gu.se/en/quality-government/qog-data/data-downloads>.

Dataset<sup>2</sup>, the 2023 Maddison Project Database<sup>3</sup>, the World Development Indicators<sup>4</sup>, the World Inequality Database<sup>5</sup> and the 13<sup>th</sup> version of the Varieties of Democracy Dataset<sup>6</sup> were all used. The data used for analysis would range from the years of 1972-2014, hence spanning over 4 decades and was measured with country-year as the unit of analysis.

Within all the statistical techniques utilized to investigate the question ‘does modernization impact democratization’, modernization was measured through logged GDP per capita, and regime type was measured by the average imputed Polity and Freedom House score. GDP per capita was deliberately logged and used as the variable `log_gdppc`<sup>7</sup> through all stages of analysis to reduce the skew present in regular GDP per capita interpretations and ensure a proportional interpretation of income in different countries. Furthermore, other data cleaning measures were adopted to ensure appropriate statistical analysis, such as the creation of `treat_gdp`<sup>8</sup>, as a treatment variable for high vs low GDP per capita countries for matching. Alongside this, the variable `log_gdppc_lag5`<sup>9</sup> was created to represent a 5-year lagged GDP per capita, which was utilized as an instrumental variable during analysis. The final step of data cleaning undertaken was filtering the variables used to not include any missing values.

Imputed Polity scores are used as a measurement for democratization, as the variable utilizes Polity scores, and imputes any missing values with those of Freedom House, to reduce any sample bias. Both indicators are commonly accepted and used in democratization literature due to them compiling various facets of regime measurements into a comprehensive and simplified points system. However, it must be acknowledged that both indicators are flawed with the Polity score having a coding bias for American standards of democracy, and with the Freedom

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<sup>2</sup> Authoritarian Regimes Dataset (2018). *Authoritarian Regimes Dataset*. [online] Google.com. Available at: <https://sites.google.com/site/authoritarianregimedataset/introduction> [Accessed 31 Dec. 2025].

<sup>3</sup> Bolt, J. and van Zanden, J. (2024). *Maddison Project Database 2023*. [online] University of Groningen. Available at: <https://www.rug.nl/ggdc/historicaldevelopment/maddison/releases/maddison-project-database-2023>.

<sup>4</sup> World Bank (2025). *World Development Indicators*. [online] The World Bank. Available at: <https://databank.worldbank.org/source/world-development-indicators>.

<sup>5</sup> World Inequality Database (2024). *Data – WID – World Inequality Database*. [online] WID - World Inequality Database. Available at: <https://wid.world/data/>.

<sup>6</sup> V-Dem (2022). *The V-Dem Dataset – V-Dem*. [online] V-dem.net. Available at: <https://v-dem.net/data/the-v-dem-dataset/>.

<sup>7</sup> See Appendix A, Section A.1 for full code

<sup>8</sup> See Appendix A, Section A.4 for full code

<sup>9</sup> See Appendix A, Section A.3 for full code

House indicator utilizing an aggregation rule that converts subscales into an overall democracy score that fails to meet the assumptions of an ordinal scale (Seawright, 2014).

The utilization of GDP per capita as a measurement of modernization has been commonly accepted in antecedent literature and analysis surrounding the topic due to economic growth being a core facet of modernization, and GDP per capita being an easily accessible measure of economic growth. However, economic growth is not always synonymous with improved well-being (Costanza, 2009, p.1), which is another facet of modernization. Hence in isolation GDP per capita is not a perfect measure for modernization and requires the implementation of other control variables when running statistical analyses.

## Empirical results

### Ordinary Least Squares Regression

To begin with, an ordinary least squares (OLS) regression is carried out to analyze the relationship between modernization, as measured through log GDP per capita, on democratization, as measured through an imputed polity score.

| OLS Regressions: Modernisation and Democracy             |                |                |
|----------------------------------------------------------|----------------|----------------|
|                                                          | Naïve OLS      | Extended OLS   |
| Log GDP per Capita                                       | 1.51*          | 0.12           |
|                                                          | [ 1.45; 1.58]  | [-0.04; 0.28]  |
| Urbanisation                                             |                | 0.01           |
|                                                          |                | [-0.00; 0.01]  |
| Female School Enrolment                                  |                | 0.03*          |
|                                                          |                | [ 0.02; 0.03]  |
| Top 10% Income Share                                     |                | -5.90*         |
|                                                          |                | [-6.82; -4.98] |
| Public Sector Corruption                                 |                | -5.09*         |
|                                                          |                | [-5.46; -4.72] |
| Intercept                                                | -7.70*         | 7.02*          |
|                                                          | [-8.24; -7.16] | [ 5.62; 8.42]  |
| R <sup>2</sup>                                           | 0.26           | 0.47           |
| Adj. R <sup>2</sup>                                      | 0.26           | 0.47           |
| Num. obs.                                                | 6485           | 4330           |
| RMSE                                                     | 2.99           | 2.49           |
| * Null hypothesis value outside the confidence interval. |                |                |

Table 1- OLS Regression

A naïve OLS was run in the beginning, solely to ensure that the correlation between modernization and democratization was statistically significant at a baseline level, and to ensure that there was a positive correlation between log GDP per capita and the imputed polity score. However, it is to be acknowledged that the naïve model suffers from omitted variable bias and hence is not extensively credible for analysis. The extended OLS helps account for this bias by including control variables that were also associated as parts of modernization, such as urbanization (Schnaiberg, 1971), female school enrolment (Yadav, 2014), the income share held by the top 10% (Inglehart, 2016) and corruption present in public sector workers (Huntington, 2002). Upon accounting for other variables, we notice that the coefficient for log GDP per capita reduces in size significantly, going from 1.51 to 0.12 and that it loses its significance. This implies that the naïve OLS model suffered from correlations that were inflated by other factors, which the Extended OLS model accounts for it appropriately. The extended OLS model has an adjusted  $R^2$  score of 0.47, which is 0.21 higher than the naïve model's, proving that while it doesn't account for all of the variation present in the model, the inclusion of controls improves model fit.

### Mediation Analysis

Given the observed decrease in the GDP per capita's coefficient and significance in the OLS model, while the control variables remain significant, this pattern suggests that modernization may influence democratization indirectly through certain mechanisms. Hence to further investigate the relationship between modernization and democratization, a mediation analysis is conducted, with female education being utilized as a mediating variable.

| Effect                 | Estimate  | CI Lower   | CI Upper  | p_value |
|------------------------|-----------|------------|-----------|---------|
| ACME (Indirect effect) | 0.2886213 | 0.2402931  | 0.3376916 | <0.001  |
| ADE (Direct effect)    | 0.1185414 | -0.0318902 | 0.2757115 | 0.122   |
| Total Effect           | 0.4071627 | 0.2633998  | 0.5546815 | <0.001  |
| Proportion Mediated    | 0.7088598 | 0.4958320  | 1.1177980 | <0.001  |

Table 2- Mediation Analysis

After conducting the mediation analysis, it can be observed that GDP per capita's direct effect on the imputed polity scores is 0.119, which remains in line the extended OLS model's estimates of 0.12, and that the direct effect is insignificant, this suggests that female education may account for a substantial proportion of GDP per capita's effect on democratization. This idea is reinforced by looking at the indirect or mediated effect, which is 0.289 and is significant, implying that when GDP per capita increases, it improves democracy scores primarily through increased female educational access. This provides evidence that female education acts as a mechanism that works to illustrate one mechanism of how GDP per capita impacts imputed polity scores. This claim is strengthened by the proportion mediated within this model being 0.709, or around 71%, indicating that 71% of GDP per capita's effect on democratization is explained through the percentage of female students who attained education.

### Instrumental Variable

While mediation analysis helps us identify a causal pathway, it alone does not account for the problem of reverse causality. Although the previous two methods suggest a relationship between modernization and democratization, they do not help rule out the possibility of democratization impacting modernization. To address this concern, Instrumental Variable (IV) analysis is undertaken, with a 5-year lagged GDP per capita being used as the instrument for the current GDP per capita. This instrument was chosen as past GDP per capita is assumed to be strongly correlated with current income and with the inclusion of the same controls utilized in the OLS regression, would not impact current democratic outcomes unless through current GDP per capita.

In the first stage regression<sup>10</sup>, it can be seen that the lagged GDP per capita has a coefficient of 0.992 and is statistically significant, reflecting that income stays persistent over time and that wealthier countries tend to stay wealthy. In addition to this, the F-statistic of this model is 6.492e+04, hence exceeding the threshold of 10 for strong instruments, reinforcing the idea that lagged GDP per capita is a strong predictor of current GDP per capita. The first stage regression thus shows that lagged GDP per capita satisfies the relevance condition.

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<sup>10</sup> See Appendix B, Section B.1 for results

Lagged GDP per capita should meet the exogeneity assumption as current democratic outcomes cannot impact past income levels. However, it should be acknowledged that past political institutions may have influenced past income levels, hence exogeneity cannot be guaranteed. Lagged GDP per capita is also assumed to meet the exclusion restriction as income levels of the past are unlikely to directly shape current democratization, unless through the means of economic development. Despite this, due to institutional persistence, this assumption cannot be tested directly and can be a limitation of the IV approach. Hence, some caution must be exercised when interpreting the IV results from the second regression.

| IV Regression: GDP per Capita → Polity Score |                            |
|----------------------------------------------|----------------------------|
|                                              | Dependent variable:        |
|                                              | Polity Score               |
| Log GDP per Capita (IV)                      | 0.502*** (0.377, 0.627)    |
| Urbanisation                                 | -0.001 (-0.007, 0.004)     |
| Top 10% Income Share                         | -4.281*** (-5.053, -3.508) |
| Public Sector Corruption                     | -4.970*** (-5.275, -4.666) |
| Constant                                     | 5.778*** (4.682, 6.874)    |
| Observations                                 | 5,163                      |
| R <sup>2</sup>                               | 0.414                      |
| Adjusted R <sup>2</sup>                      | 0.413                      |
| Residual Std. Error                          | 2.620 (df = 5158)          |
| Note:                                        | *p**p***p<0.01             |

Table 3- Instrumental Variable

Upon conducting the second stage regression, we notice that the GDP per capita has a positive coefficient of 0.502 which is statistically significant, this supports our hypothesis that modernization does have a positive impact on democratization. The IV estimate of 0.502 is also much larger than the Extended OLS model's estimate of 0.12, implying that the OLS model may be biased downwards due to measurement errors, and due to IV isolating and looking at only the exogenous variation in GDP per capita.

### Matching

With the IV regression aiding in establishing causal credibility between modernization and democratization, matching will be utilized to ensure that the relationship is robust to selection bias as well. As there is no established treatment that countries undergo to democratize, a



variable called `treat_gdp`, which considered countries with a GDP per capita higher than the world's median to be 'treated' and those with GDP per capita's that were lower to be the 'control'.

Upon creating the matched dataset, the standard mean differences of all the variables were much closer to 0<sup>11</sup> and were within conventional thresholds, proving that matching helped improve covariate balance between the treated and control group.

|                                                                                                                 | Matched Sample      |
|-----------------------------------------------------------------------------------------------------------------|---------------------|
| (Intercept)                                                                                                     | 4.913***<br>(0.111) |
| treat_gdp                                                                                                       | 1.945***<br>(0.165) |
| Num.Obs.                                                                                                        | 1410                |
| R2                                                                                                              | 0.090               |
| R2 Adj.                                                                                                         | 0.089               |
| AIC                                                                                                             | 7193.7              |
| BIC                                                                                                             | 7209.5              |
| RMSE                                                                                                            | 3.10                |
| <ul style="list-style-type: none"> <li>• p &lt; 0.1, * p &lt; 0.05, ** p &lt; 0.01, *** p &lt; 0.001</li> </ul> |                     |

Table 4- Matching

Upon conducting regressions with the matching estimates, it can be seen that within this matched sample, countries with a higher GDP per capita than the world median have democracy scores 1.945 points higher than low-income countries. This reinforces the positive relationship the Extended OLS and the IV have also found between modernization and democratization, suggesting that the result is robust.

## Conclusion

In conclusion, this essay examined whether modernization, measured through GDP per capita, impacts democratization, represented by an imputed Polity score. All 4 quantitative techniques utilized were largely consistent with the hypothesis and suggested a positive relationship between the two variables, while also providing further insights. The results indicated that economic growth impacted democracy largely through indirect methods, such as female education. With the GDP per capita variable losing its significance in the extended OLS model,

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<sup>11</sup> See Appendix B, Section B.2 for results

it suggests that modernization does not directly cause democracy but causes it through social and institutional mechanisms that it helps develop. The findings of this report are that while economic growth does contribute to democratization, it should be accompanied with social change, such as increased education, with the addition of control variables which measure corruption and inequality also impacting democratization to a large extent. Future research could look towards investigating more mediating mechanisms and looking into whether democratization trends differ based on regions.

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## **Appendix-**

### **Appendix A: Code**

#### **A1- OLS code**

```
QOG <- read.csv("qog_std_ts_jan25.csv")
```

```
QOG$log_gdppc <- log(QOG$mad_gdppc)
```

```
install.packages("skimr")
```

```
library(skimr)
```

```
skim(QOG[, c("ht_ipolity2",  
            "log_gdppc",  
            "wdi_popurb",  
            "wdi_gerp",  
            "top_top10_income_share",  
            "vdem_pubcorr"))]
```

```
sum(complete.cases(QOG[,  
c('ht_ipolity2','log_gdppc','wdi_popurb','wdi_gerp','top_top10_income_share','vdem_pubcorr'  
'))])
```

```
library(estimatr)
```

```
library(texreg)
```

```
m_ols_naive <- lm_robust(ht_ipolity2 ~ log_gdppc, data = QOG)
```

```
summary(m_ols_naive)
```

```
m_ols_extended <- lm_robust(
  ht_ipolity2 ~ log_gdppc + wdi_popurb + wdi_gerpfp + top_top10_income_share
+ vdem_pubcorr,
  data = QOG
)
```

```
summary(m_ols_extended)
```

```
htmlreg(
  list(m_ols_naive, m_ols_extended),
  file = "OLS_comparison.doc",
  doctype= FALSE,
  custom.model.names = c("Naive OLS", "Extended OLS"),
  custom.coef.map = list(
    'log_gdppc' = "Log GDP per Capita",
    'wdi_popurb' = "Urbanisation",
    'wdi_gerpfp' = "Female School Enrollment",
    'top_top10_income_share' = "Top 10% Income Share",
    'vdem_pubcorr' = "Public Sector Corruption",
    '(Intercept)' = "Intercept"
  ),
  caption = "OLS Regressions : Modernisation and Democracy",
  caption.above = TRUE
)
```

## **A2- Mediation Analysis code**

```
QOG <- read.csv("qog_std_ts_jan25.csv")
```

```
QOG$log_gdppc <- log(QOG$mad_gdppc)
```

```
install.packages("mediation")
```

```
install.packages("sandwich")
```

```
install.packages("lmtest")
```

```
library(mediation)
```

```
library(sandwich)
```

```
library(lmtest)
```

```
library(tidyverse)
```

```
med_data <- QOG %>%
```

```
  dplyr::select(
```

```
    ht_ipolity2,
```

```
    log_gdppc,
```

```
    wdi_gerpf,
```

```
    wdi_popurb,
```

```
    top_top10_income_share,
```

```
    vdem_pubcorr
```

```
  ) %>%
```

```
  na.omit()
```

```
model.m <- lm(
```

```
  wdi_gerpf ~ log_gdppc + wdi_popurb + top_top10_income_share + vdem_pubcorr,
```

```
  data = med_data
```

```
)
```

```
summary(model.m)
```

```
model.y <- lm(
```

```
  ht_ipolity2 ~ log_gdppc + wdi_gerpf + wdi_popurb +
```

```
  top_top10_income_share + vdem_pubcorr,
```

```
  data = med_data
```

```
)
```

```
summary(model.y)
```

```

med.out <- mediate(
  model.m,
  model.y,
  treat = "log_gdppc",
  mediator = "wdi_gerpf",
  sims = 1000,
  boot = TRUE
)

summary(med.out)

install.packages("dplyr")
install.packages("knitr")

library(dplyr)
library(knitr)
library(kableExtra)

med_table <- data.frame(
  Effect = c("ACME (Indirect effect)", "ADE (Direct effect)", "Total Effect", "Proportion
Mediated"),
  Estimate = c(med.out$d0, med.out$z0, med.out$tau.coef, med.out$n0),
  CI_Lower = c(med.out$d0.ci[1], med.out$z0.ci[1], med.out$tau.ci[1], med.out$n0.ci[1]),
  CI_Upper = c(med.out$d0.ci[2], med.out$z0.ci[2], med.out$tau.ci[2], med.out$n0.ci[2]),
  p_value = c(med.out$d0.p, med.out$z0.p, med.out$tau.p, med.out$n0.p)
)

# Print table
kable(med_table, caption = "Mediation Analysis: GDP per Capita → Female Education →
Democracy")

```

### A3- Instrumental Variable code

```
QOG <- read.csv("qog_std_ts_jan25.csv")
```

```
QOG$log_gdppc <- log(QOG$mad_gdppc)
```

```
install.packages("ivreg", dependencies = TRUE)
```

```
library(ivreg)
```

```
library(dplyr)
```

```
QOG <- QOG %>%
```

```
  arrange(cname_qog, year) %>%
```

```
  group_by(cname_qog) %>%
```

```
  mutate(log_gdppc_lag5 = lag(log_gdppc, 5)) %>%
```

```
  ungroup()
```

```
med_data <- QOG %>%
```

```
  filter(!is.na(log_gdppc) & !is.na(log_gdppc_lag5) &
```

```
         !is.na(ht_ipolity2) & !is.na(wdi_popurb) &
```

```
         !is.na(top_top10_income_share) & !is.na(vdem_pubcorr))
```

```
first_stage <- lm(log_gdppc ~ log_gdppc_lag5 + wdi_popurb +
```

```
                 top_top10_income_share + vdem_pubcorr, data = QOG)
```

```
summary(first_stage)
```

```
iv_model <- ivreg(ht_ipolity2 ~ log_gdppc + wdi_popurb +
```

```
                 top_top10_income_share + vdem_pubcorr |
```

```
                 log_gdppc_lag5 + wdi_popurb +
```

```
                 top_top10_income_share + vdem_pubcorr,
```

```
                 data = QOG)
```

```
summary(iv_model, diagnostics = TRUE)
```

```
install.packages("stargazer")
```



```
library(stargazer)

stargazer(iv_model,
  type = "html",
  title = "IV Regression: GDP per Capita → Polity Score",
  dep.var.labels = "Polity Score",
  covariate.labels = c("Log GDP per Capita (IV)",
    "Urbanisation",
    "Top 10% Income Share",
    "Public Sector Corruption"),
  digits = 3,
  ci = TRUE,
  ci.level = 0.95,
  single.row = TRUE,
  out = "IV_only_results.doc")
```

#### **A4- Matching code**

```
QOG <- read.csv("qog_std_ts_jan25.csv")

QOG$log_gdppc <- log(QOG$mad_gdppc)

install.packages(c("MatchIt", "cobalt", "estimatr", "tidyverse", "kableExtra"))

library(MatchIt)
library(cobalt)
library(estimatr)
library(tidyverse)
library(kableExtra)

QOG <- QOG %>%
  mutate(treat_gdp = ifelse(log_gdppc > median(log_gdppc, na.rm=TRUE), 1, 0))

QOG_clean <- QOG %>%
```

```
filter(!is.na(treat_gdp),
       !is.na(wdi_popurb),
       !is.na(top_top10_income_share),
       !is.na(wdi_gerp))
```

```
psm_lit <- matchit(
  treat_gdp ~ wdi_popurb + top_top10_income_share + wdi_gerp,
  data = QOG_clean,
  method = "nearest",
  distance = "logit",
  caliper = 0.2
)
```

```
summary(psm_lit)
```

```
love.plot(
  psm_lit,
  stats = "mean.diffs",
  threshold = 0.1,
  abs = TRUE
)
```

```
match_df <- match.data(psm_lit)
```

```
aggregate(distance ~ treat_gdp, data = match_df, mean)
```

```
model_match <- lm_robust(
  ht_ipolity2 ~ treat_gdp,
  data = match_df,
  weights = weights
)
```

```
summary(model_match)
```

```
library(modelsummary)
```

```
out_lm <- lm_robust(  
  ht_ipolity2 ~ treat_gdp,  
  data = match_df,  
  weights = weights  
)  
install.packages("pandoc")  
library(pandoc)
```

```
modelsummary(  
  list("Matched Sample" = out_lm),  
  stars = TRUE,  
  statistic = "std.error",  
  output = "matched_results.docx"  
)
```

## **Appendix B: Supplementary Results**

## B1- First stage regression for Instrumental variable

```
Call:
lm(formula = log_gdppc ~ log_gdppc_lag5 + wdi_popurb + top_top10_income_share +
    vdem_pubcorr, data = Q0G)

Residuals:
    Min       1Q   Median       3Q      Max
-2.08858 -0.08327  0.00070  0.09360  1.47833

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    0.2127888   0.0343224   6.200 6.00e-10 ***
log_gdppc_lag5  0.9916682   0.0039064 253.856 < 2e-16 ***
wdi_popurb      0.0007054   0.0001744   4.044 5.32e-05 ***
top_top10_income_share -0.1732798   0.0259878  -6.668 2.81e-11 ***
vdem_pubcorr    -0.0032782   0.0100159  -0.327   0.743
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1912 on 6536 degrees of freedom
(5850 observations deleted due to missingness)
Multiple R-squared:  0.9755,    Adjusted R-squared:  0.9754
F-statistic: 6.492e+04 on 4 and 6536 DF,  p-value: < 2.2e-16
```

## B2- Creation of matched dataset for matching

```
Call:
matchit(formula = treat_gdp ~ wdi_popurb + top_top10_income_share +
    wdi_gerp, data = Q0G_clean, method = "nearest", distance = "logit",
    caliper = 0.2)

Summary of Balance for All Data:
              Means Treated Means Control Std. Mean Diff. Var. Ratio eCDF Mean eCDF Max
distance              0.8312           0.2455           2.5715    0.7565    0.4338    0.7247
wdi_popurb            68.9817          34.5396           2.0945    1.1125    0.4276    0.7032
top_top10_income_share  0.4311           0.5033          -0.6408    2.2499    0.1924    0.4057
wdi_gerp             102.7791          92.5224           1.1702    0.1141    0.1592    0.3570

Summary of Balance for Matched Data:
              Means Treated Means Control Std. Mean Diff. Var. Ratio eCDF Mean eCDF Max Std. Pair Dist.
distance              0.5537           0.5030           0.2226    1.2053    0.0476    0.1185    0.2227
wdi_popurb            52.4531          48.5601           0.2367    1.7120    0.0476    0.1140    0.3987
top_top10_income_share  0.4850           0.4927          -0.0685    2.3596    0.0699    0.1490    0.8410
wdi_gerp             102.3295          101.9137           0.0474    0.3375    0.0851    0.1603    1.6493

Sample Sizes:
              Control Treated
All             2325    3382
Matched          886     886
Unmatched        1439    2496
Discarded         0         0
```